

SOP-110 — Rotating Equipment Vibration Monitoring and Bearing Lubrication

Plant Site A · Rev 5 · Issued 22/01/2025 · Approved by Plant Engineering

1. Purpose

This procedure defines the routine vibration monitoring programme and the bearing lubrication practice for rotating equipment at Plant Site A. Rising vibration is the earliest practical warning of bearing distress, misalignment, unbalance and cavitation, and a disciplined monthly route converts sudden failures into planned work.

2. Scope

Applies to all pumps in the equipment register, including the process transfer pumps and the utilities cooling water pumps such as P-102A, together with their electric motor drivers. Machines with permanently installed online monitoring are read from the system but still receive the manual route for confirmation.

3. References

ISO 10816 vibration severity guidance, held in the technical library.

SOP-101 Centrifugal Pump Startup and Shutdown Procedure.

Lubrication schedule and approved lubricants list for Plant Site A.

4. Safety Precautions

Take readings with the machine at normal steady duty. Do not reach past rotating elements to place the probe; every measurement point is accessible outside the guard.

Wear hearing protection in the pump bay. If a machine sounds audibly distressed, do not linger to complete the route; report it immediately.

When greasing with the machine running, use the extension hose and stand clear of the coupling plane.

5. Tools and Materials

Have available: the portable vibration analyser with accelerometer and magnetic mount, charged and loaded with the current month's route, the stethoscope probe, the grease gun charged with the approved grade for the machines on the route, clean rags, the lubrication schedule, and hearing protection. Verify the analyser against the reference check source before starting the route each month.

Use the same marked measurement point every month. Paint dot each bearing position; readings taken a few centimetres apart on the same casing differ enough to corrupt the trend and mask genuine change.

6. Procedure

6.1 Follow the monthly route in the route list order. At each machine, measure vibration velocity in millimetres per second RMS at the drive end and non-drive end bearings, in horizontal, vertical and axial directions.

6.2 Take readings only at steady normal duty. A pump throttled for process reasons reads differently, and a reading captured during transient operation shall be flagged or retaken rather than entered silently into the trend.

6.3 Record each reading against the machine tag and point identifier in the route logger. Never round or estimate; the trend matters more than the single value.

6.4 Compare each reading against the ISO 10816 zone boundaries for the machine class. Readings in zone C require a work request within one week; readings in zone D require the machine to be reported for shutdown assessment the same shift.

6.5 Compare the reading against the previous three months. A rise of more than fifty percent over baseline is reportable even if the absolute value remains inside zone B, because rate of change is the stronger failure signal.

6.6 Where a reading is elevated, capture a spectrum. Bearing defect frequencies, running speed harmonics from misalignment, and blade pass energy from cavitation each present distinctly, and the spectrum directs the correct maintenance response.

6.7 Where cavitation is suspected from the spectrum or from the audible gravel noise, check the suction strainer differential pressure per SOP-114 before condemning the pump hydraulics. A choked strainer reproduces every classic cavitation symptom and is corrected in an hour.

6.8 Listen to each bearing with the stethoscope probe. A dry rumble that varies with speed indicates lubrication starvation and shall be corrected the same day.

6.9 Grease motor bearings at the interval and quantity shown in the lubrication schedule using only the approved grease grade. Wipe the fitting before connecting, and grease slowly with the relief port open until clean grease appears.

6.10 Do not over-grease. More motor bearings fail from churning excess grease than from starvation. If the relief port does not purge, stop and raise a work request rather than forcing additional grease.

6.11 For two pump sets with one unit on standby, coordinate with operations to capture the standby unit during its scheduled exercise run, so that both units of the set carry current readings.

6.12 Check oil-lubricated pump bearing housings at the sight glass, top up to the centre mark if required, and record any housing that needed more than a token top-up as a potential leak for follow-up.

6.13 Report any machine with repeated elevated readings on the same measurement point over consecutive months to the reliability engineer for root cause investigation. Repeated symptoms with no root cause action is how chronic failures are made.

7. Acceptance Criteria

The route is complete when every machine on the list has a full set of readings recorded, every zone C or D finding has a work request raised with the reading attached, and the lubrication tasks due in the month are signed off in the schedule.

8. Records

Route readings are retained in the vibration database against each machine tag. Work requests raised from the route shall quote the reading, the zone, and the trend over the previous three months so that maintenance can prioritise on evidence. The route summary and the list of machines in zone C or above shall be tabled at the monthly reliability meeting, each with its open work order number.